

Exercise 3

CASE Statements

(1) Table products

Product_id	Product_name	Price
1	Laptop	1200.00
2	Phone	800.00
3	Keyboard	48.00
4	Monitor	300.00
5	Mouse	25.00

1) SELECT ~~DISTINCT~~ Price, Product_name
CASE

WHEN Price BETWEEN 100 AND 1000 THEN 'Mid-range'

WHEN Price > 1000 THEN 'Expensive'

WHEN Price < 100 THEN 'Budget'

ELSE Price

END AS Price-range Price-category

FROM products

Product_name	Price	Price-category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Keyboard	48.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

2. Table orders

Order_id	customer_name	amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	48.50
5	Ethan	1200.00

customer_name
 SELECT ~~DISTINCT~~ amount
 CASE

WHEN amount $\geq$ 1000 THEN 'High value'

WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium value'

WHEN amount $<$ 500 THEN 'Low value'

ELSE amount

END AS Order_value_category

Results: FROM Orders

Customer-name	amount	order_value_category
1 Alice	150.00	Low Value
2 Bob	560.00	Medium value
3 Charlie	999.99	Medium value
4 Diana	45.50	Low value
5 Ethan	12000	High Value

3. Table employees

Emp-id	emp-name	Department	Salary
1	John	IT	85000
2	Sara	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

SELECT emp-name, department, salary

WHEN department $\neq$ IN 'IT' AND salary $>$ 80000

THEN 'Senior IT'

WHEN department IN 'HR' AND salary $>$ 55000 THEN 'Experienced HR'

ELSE 'Staff'

END AS position-level FROM employees

3)

emp_name	Department	Salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

(1) Table Students

Student_id	Student_name	Score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

14). SELECT Student_name,
Score

CASE 1:

WHEN Score $\geq$ 90 THEN 'A'

WHEN Score BETWEEN 80 AND 89 THEN 'B'

WHEN Score BETWEEN 70 AND 79 THEN 'C'

WHEN Score BETWEEN 60 AND 69 THEN 'D'

ELSE 'F'

END AS Grade

FROM Students

Student_name	Score	Grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

6. SELECT issue-type
 priority
 CASE
 WHEN priority = 3 THEN 'High'
 WHEN priority = 2 THEN 'Medium'
 ELSE Low
 END AS priority-label
 FROM tickets

issue-type	priority	priority-label
login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7 Table attendance

student-id	days-present	total-days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

SELECT student-id
 (days-present * 100.0 / total-days) AS attendance-percentage
 CASE
 WHEN (days-present * 100.0 / total-days) >= 90 THEN 'Excellent'
 WHEN (days-present * 100.0 / total-days) BETWEEN 75 AND 89
 THEN 'Good'
 ELSE 'Needs improvement'
 END AS attendance-status

(5) TABLE deliveries

Delivery-id	delivery-time-minutes
1	45
2	80
3	30
4	65
5	100

SELECT *
 CASE

WHEN delivery-time-minutes <= 30 THEN 'Fast'
 WHEN delivery-time-minutes BETWEEN 31 AND 60 THEN 'OnTime'
 WHEN ELSE 'Late'
 END AS performance
 FROM deliveries

Delivery-id	delivery-time-minutes	performance
1	45	On-time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

6 Tickets

Ticket-id	issue-type	Priority
1	login issue	1
2	Server down	3
3	Slow system	2
4	Email error	2
5	Password reset	1

FROM products_inventory

Product_id	Stock_qty	Stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

9. Classes

class_id	Subject	enrolled_students
1	Maths	30
2	English	25
3	Science	15
4	Art	5
5	History	20

SELECT subject,
enrolled_student

CASE

WHEN enrolled_students >= 25 THEN 'Large'

WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'

ELSE 'Small'

END AS class_size_category

FROM classes

Subject	enrolled_students	class_size_category
Maths	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

FROM attendance

Student_id	attendance	
	days-present	total-days
1	48	
2	20	
3	48	
4	25	
5	50	

Student_id	attendance_percentage	attendance_status
1	90%	Excellent
2	60%	Needs improvement
3	96%	Excellent
4	50%	Needs improvement
5	100%	Excellent

8 Products_inventory

Product_id	Stock_qty
1	5
2	0
3	25
4	10
5	3

SELECT product_id,
Stock_qty,

CASE

WHEN Stock_qty = 0 THEN 'Out of Stock'

WHEN Stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'

ELSE 'In Stock'

END AS Stock_Status

10. Payments

payment_id	amount	payment-method
1	50.00	Card
2	200.00	Cash
3	150.00	Card
4	75.00	Paypal
5	300.00	Cash

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SELECT payment_id
       payment-method
       amount
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CASE

WHEN ~~for~~ payment-method = 'Cash' AND amount >= 200 THEN

'Eligible for Discount'

ELSE 'Not Eligible'

END AS discount-eligibility

FROM payments

Payment_id	payment-method	amount	discount-eligibility
1	Card	50.00	Not eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not eligible
4	PayPal	75.00	Not eligible
5	Cash	300.00	Eligible for Discount